AI-BASED SYMPTOM-TO-INFORMATION ASSISTANT USING NLP AND RETRIEVAL-AUGMENTED GENERATION

MINOR PROJECT
ABSTRACT REPORT
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PROJECT TITLE

"AI-Based Symptom-to-Information Assistant Using NLP and Retrieval-Augmented Generation"

ABSTRACT

Access to reliable and understandable health information is often delayed by long clinic queues, limited consultation time, and the difficulty many people face in describing symptoms using medical terminology. As a result, individuals frequently turn to unstructured web searches, which often return conflicting, outdated, or overly technical content. This project addresses the need for an intelligent conversational system that helps users understand their symptoms in simple language and locate credible health information, supporting informed decisions about whether professional consultation is required, without attempting to replace that consultation.

The proposed system, the AI-Based Symptom-to-Information Assistant, combines Natural Language Processing (NLP) with a Retrieval-Augmented Generation (RAG) pipeline. When a user describes their symptoms in everyday language, the NLP module extracts key symptom-related terms and, where the description is incomplete, generates relevant follow-up questions to gather additional context such as duration, severity, or associated symptoms. This refined information is used to construct a structured query, which the RAG component uses to retrieve relevant passages from a curated set of trusted health-information sources. A language model then synthesizes these retrieved passages into a clear, easy-to-understand response, keeping the output grounded in the retrieved content rather than relying solely on the model's internal knowledge.

To encourage responsible use, the system incorporates a basic risk and urgency classification step that flags symptom combinations commonly associated with the need for prompt medical attention, prompting the user to consult a healthcare professional without delay. The interface is designed to clearly separate general informational content from situations warranting urgent care.

The expected outcome of this project is a working demonstration of how NLP and RAG techniques can be combined to make preliminary health-related information more accessible and understandable to non-experts. The system, however, has clear limitations: its knowledge is restricted to its reference material, it cannot examine or monitor patients, and it does not perform any form of medical diagnosis. It is intended solely as an educational and informational tool, and users are consistently advised to consult qualified healthcare professionals for accurate diagnosis and treatment.

KEYWORDS

Artificial Intelligence, Natural Language Processing, Retrieval-Augmented Generation (RAG), Healthcare Information, Symptom Analysis, Machine Learning.

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